

Multiple Choice (10 marks)

1. Adding more basis functions in a linear model, pick the most probably option:
 - (a) Decreases model bias
 - (b) Decreases estimation bias
 - (c) Decreases variance
 - (d) Doesn't affect bias and variance
2. Statement 1| Layer Normalization is used in the original ResNet paper, not Batch Normalization. Statement 2| DCGANs use self-attention to stabilize training.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. If your training loss increases with number of epochs, which of the following could be a possible issue with the learning process?
 - (a) Regularization is too low and model is overfitting
 - (b) Regularization is too high and model is underfitting
 - (c) Step size is too large
 - (d) Step size is too small
4. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
 - (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$
5. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3

True / False (10 marks)

Answer True or False

6. True or False: Neural networks trained with gradient descent do not always converge to the global optimum, particularly in non-convex landscapes.
7. True or False: The computational complexity of gradient descent is polynomial in the number of features, denoted as D .
8. True or False: In Naive Bayes, attributes are statistically independent of one another given the class value.
9. True or False: Increasing the amount of training data will always eliminate overfitting in machine learning models.
10. True or False: ResNeXt models in 2018 predominantly utilized ReLU activation functions instead of tanh.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the potential reasons for the poor performance of a decision tree in spam classification, particularly focusing on the implications of using shallow trees.
12. Discuss how to generate a 10×5 Gaussian matrix with entries sampled from $\mathcal{N}(\mu = 5, \sigma^2 = 16)$ and a 10×10 uniform matrix from $U[-1, 1)$. Provide the commands used in PyTorch and explain the rationale behind each step.

End of Paper